

Can Small LLMs Do Epigenetics?

Benchmarking and Fine-Tuning for DNA Methylation QA

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Motivation

DNA methylation is central to gene regulation, aging, cancer, and disease risk. It acts like a chemical “switch” on DNA, helping turn nearby genes on or off without changing the genetic code. Since the first genome-wide human methylomes in 2009, the field has grown rapidly across epigenetic clocks, disease-risk prediction, cancer profiling, and lifestyle or therapeutic intervention studies.

The field’s expanding literature creates a need for **LLMs that can synthesize findings and answer specialized scientific questions.** However, frontier models are expensive, closed-source, and difficult to audit. **Small open-source models** may be more practical for research labs if they can be effectively adapted to domains like DNA methylation.

- This project asks:
1. **How well do current LLMs answer questions about DNA methylation?**
 2. **Can a small open-source model be fine-tuned to approach frontier-model performance?**

Benchmark Design

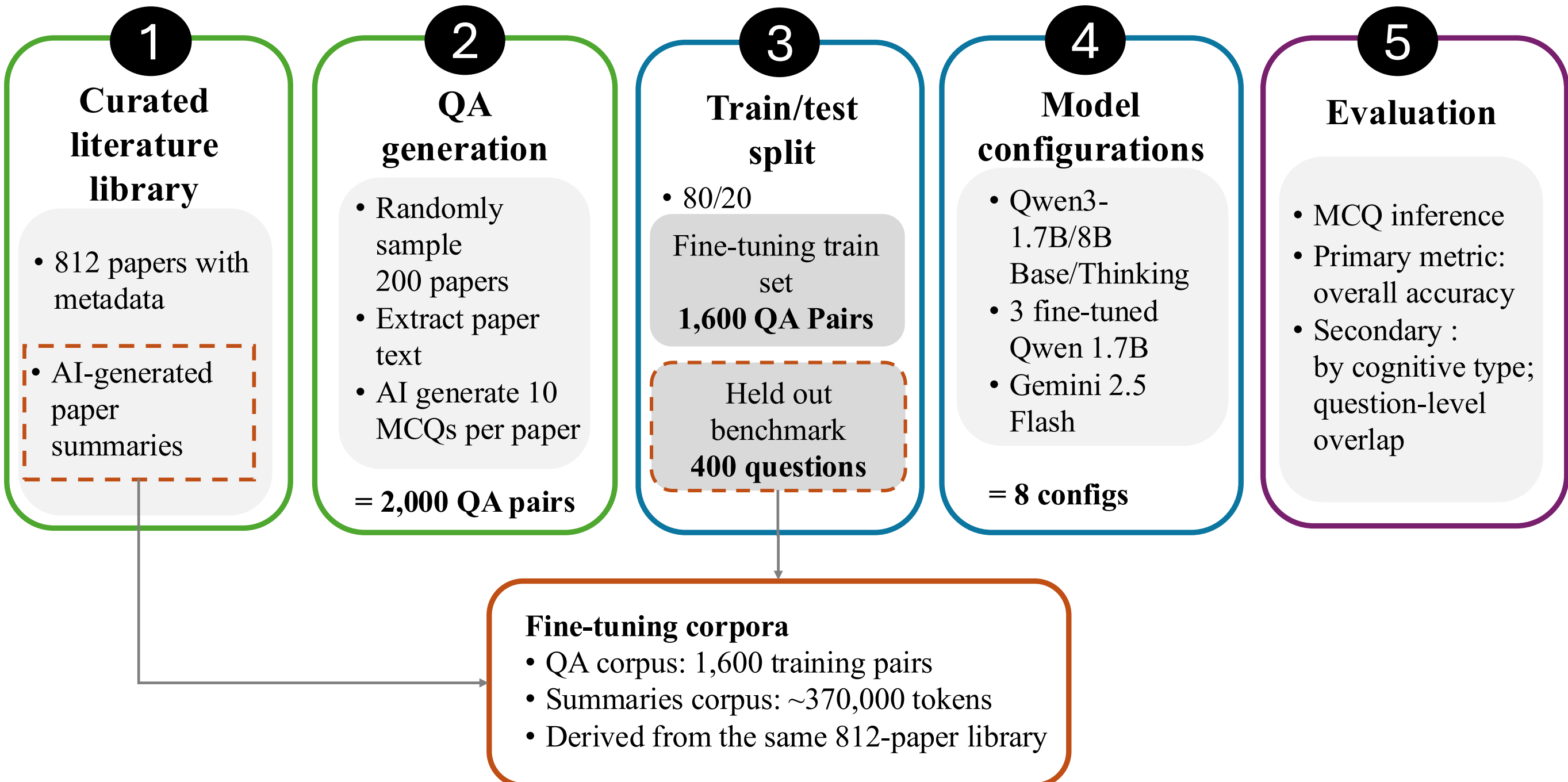
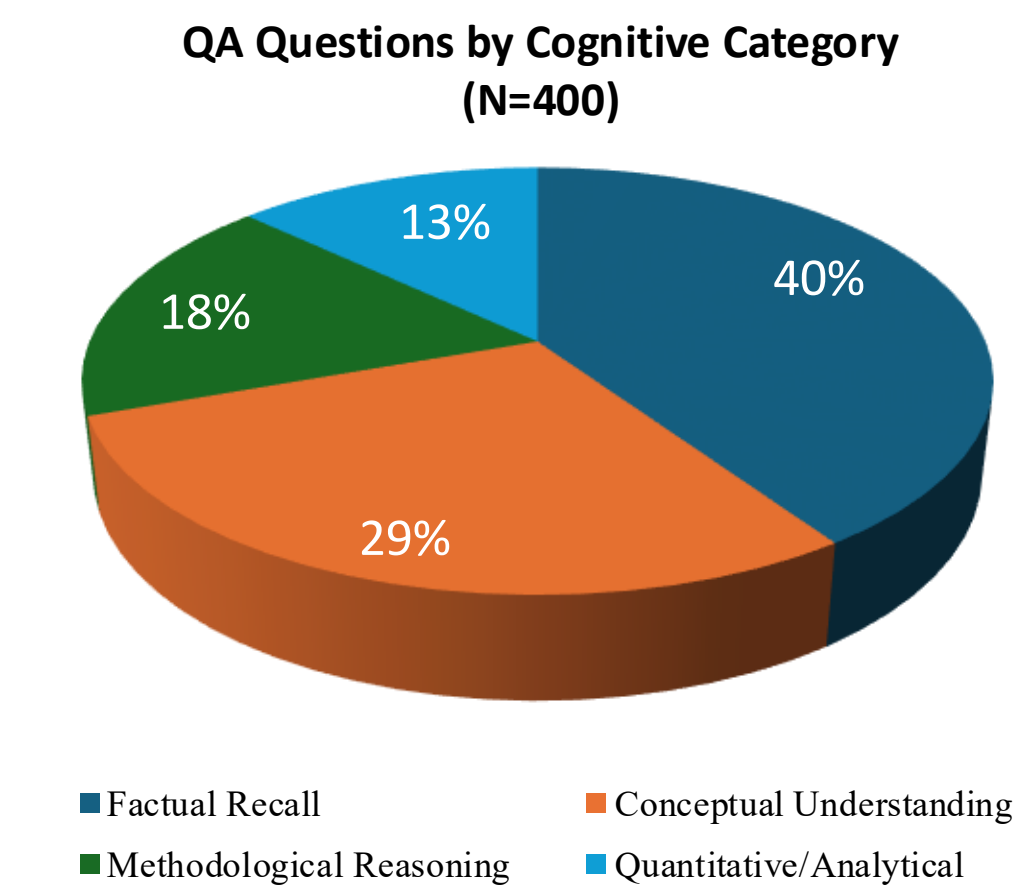


Fig 1. The eight model configurations evaluated in this study. Two-staged: continued pretraining on summaries corpus, then supervised fine-tuning on QA pairs. Single-stage: both corpora mixed in one SFT pass.



Why literature QA instead of table memorization?

An earlier version asked models to recall exact CpG identifiers from methylation tables. The fine-tuned model learned the format of CpG IDs but hallucinated incorrect identifiers. This motivated a shift toward literature-based scientific QA, where models can add value by reasoning over mechanisms, methods, and results.

Tabular Memorization QA — fine-tuned model output

Prompt: List CpG sites linked to insulin resistance.
Ground truth: The 4 CpG sites associated with insulin resistance are cg00004939, cg00007800, cg00008819, cg00017601.
Fine-tuned model prediction: The CpG site associated with insulin resistance is cg00003228. **✗ Hallucinated identifier**

Literature-Based QA — Conceptual Understanding

Q: Why does DNA methylation at CpG islands in gene promoter regions typically lead to transcriptional silencing?
A: Methylation of promoter CpG islands recruits methyl-binding domain (MBD) proteins and chromatin remodelers that promote a closed, heterochromatic state. This blocks transcription factor binding and RNA polymerase II recruitment, silencing the associated gene.

Figure 2. Sample pairs for tabular memorization-based QA vs literature-based QA that tests conceptual and mechanistic understanding.

Results & Analysis

A two-stage fine-tuned Qwen3-1.7B reaches 74.3%, which is +18.5 pts over base, within 4 pts of Gemini 2.5 Flash, and beating Qwen3-8B Thinking at less than a quarter of the parameters.

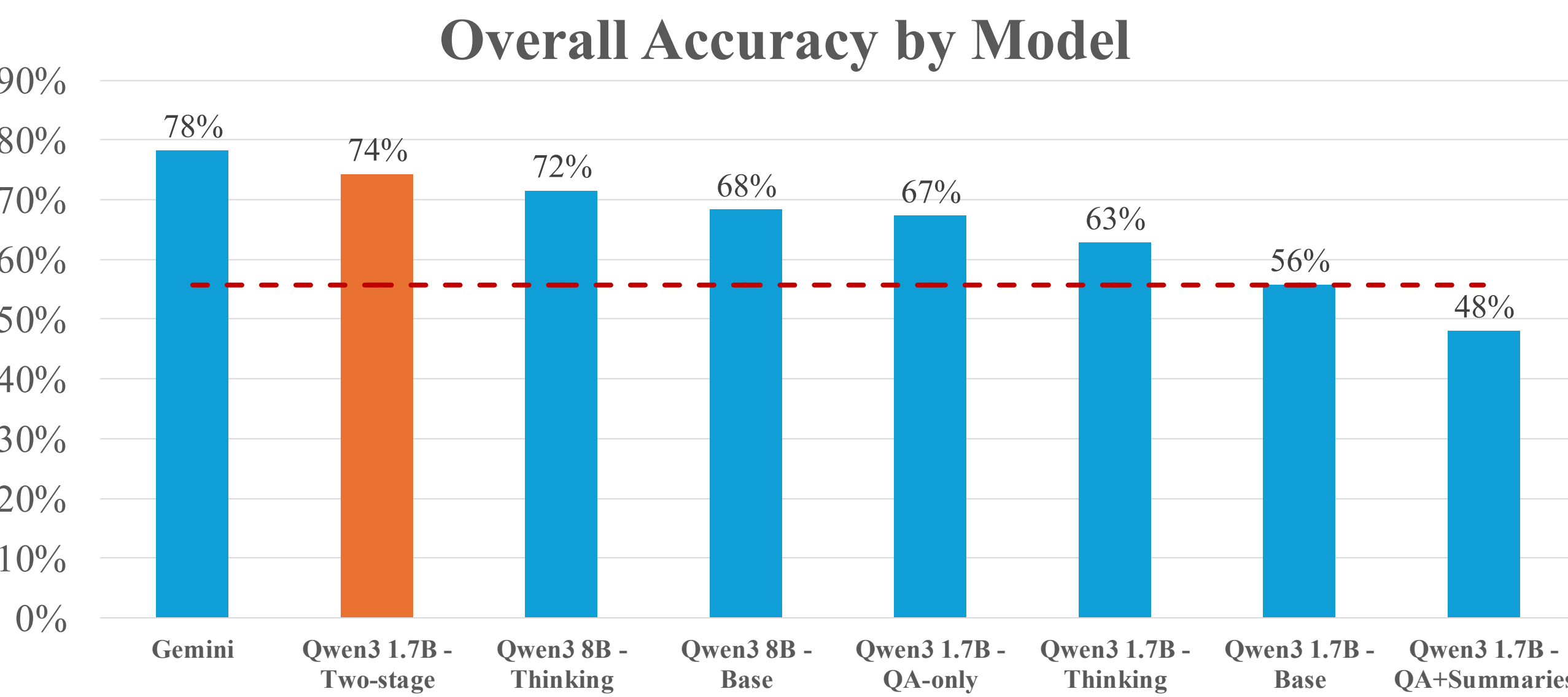


Figure 3. Overall benchmark accuracy across all eight evaluated model configurations.

Accuracy by cognitive category

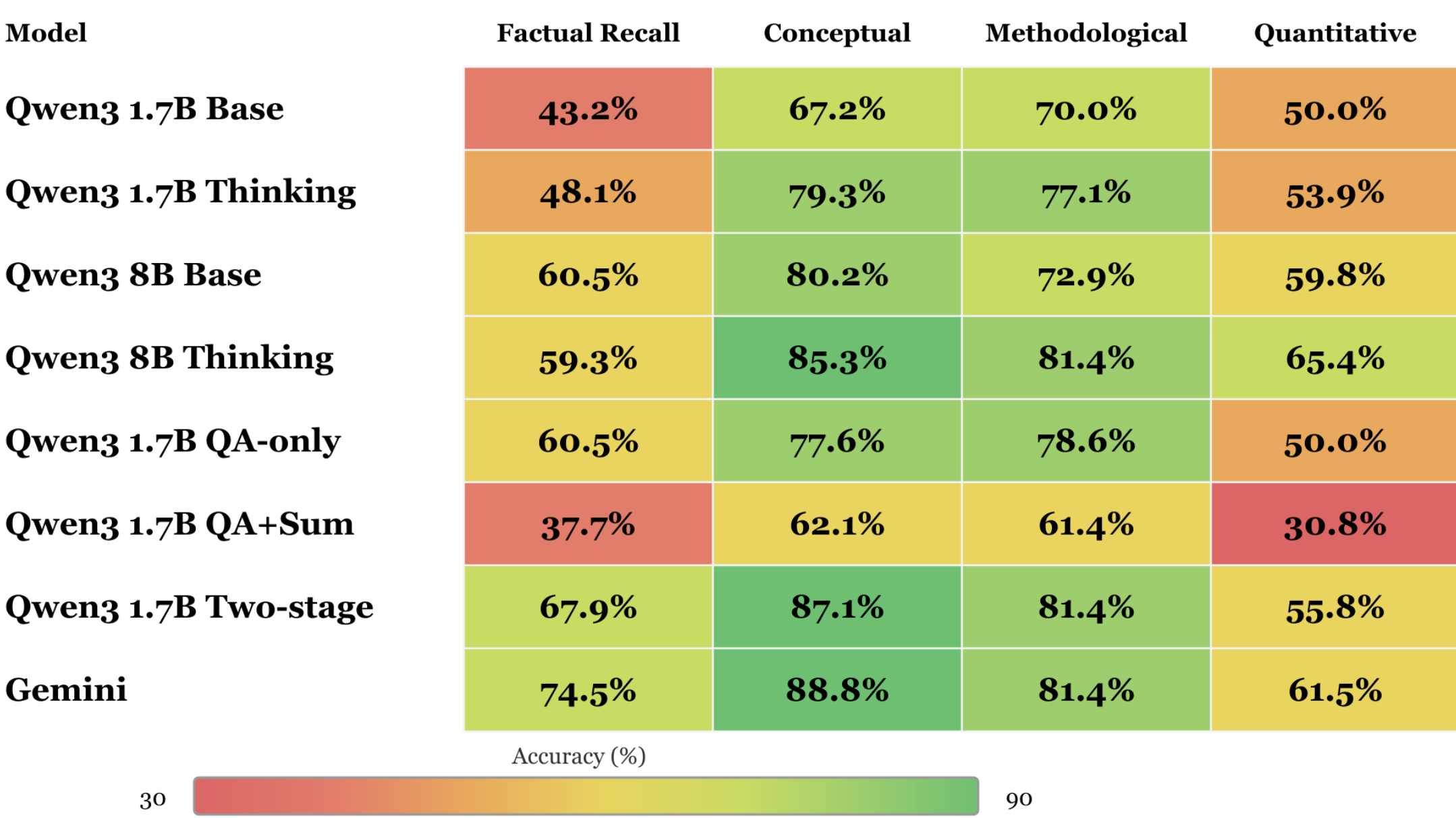


Figure 4. Heatmap showing accuracy by cognitive type across models. Two-stage Qwen3-1.7B nearly matches Gemini on conceptual and methodological questions, while factual recall and quantitative questions remain more difficult for closed-book models.

Question Level Overlap

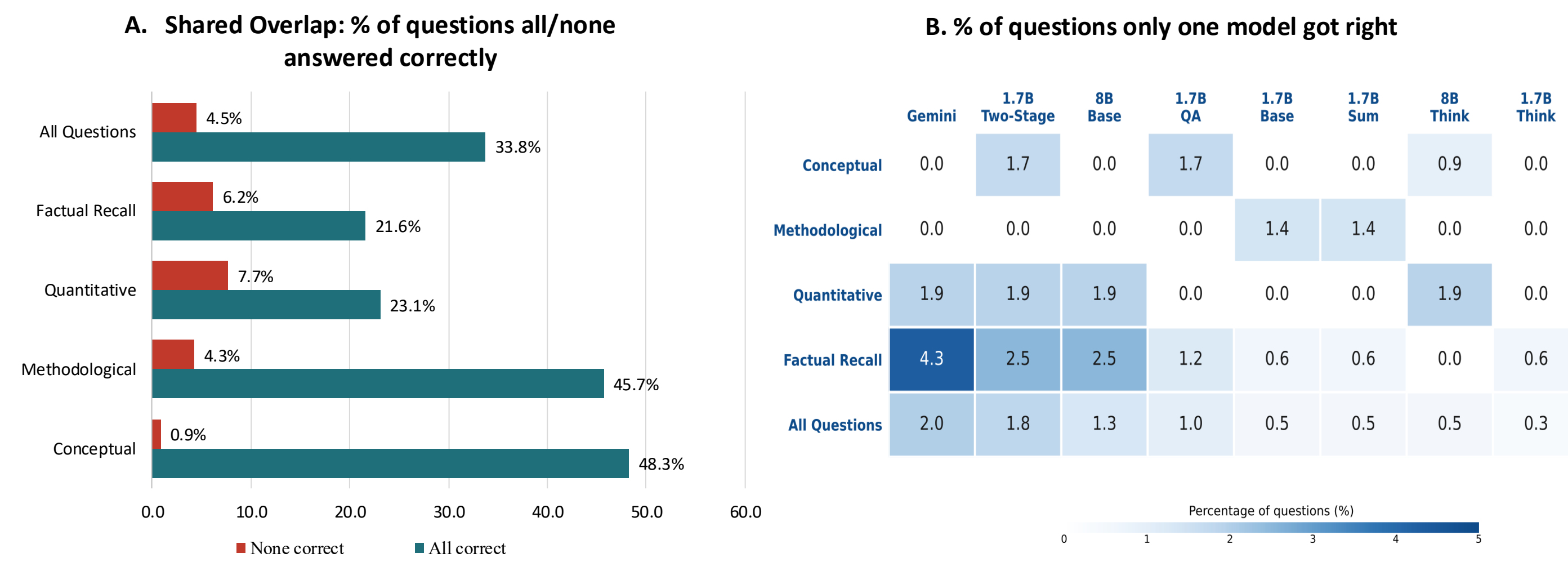


Figure 5. Question-level overlap. Conceptual and methodological questions have high shared-correct rates, while factual-recall and quantitative questions show more model-specific variation and unique model strengths.

Effect Size Decomposition

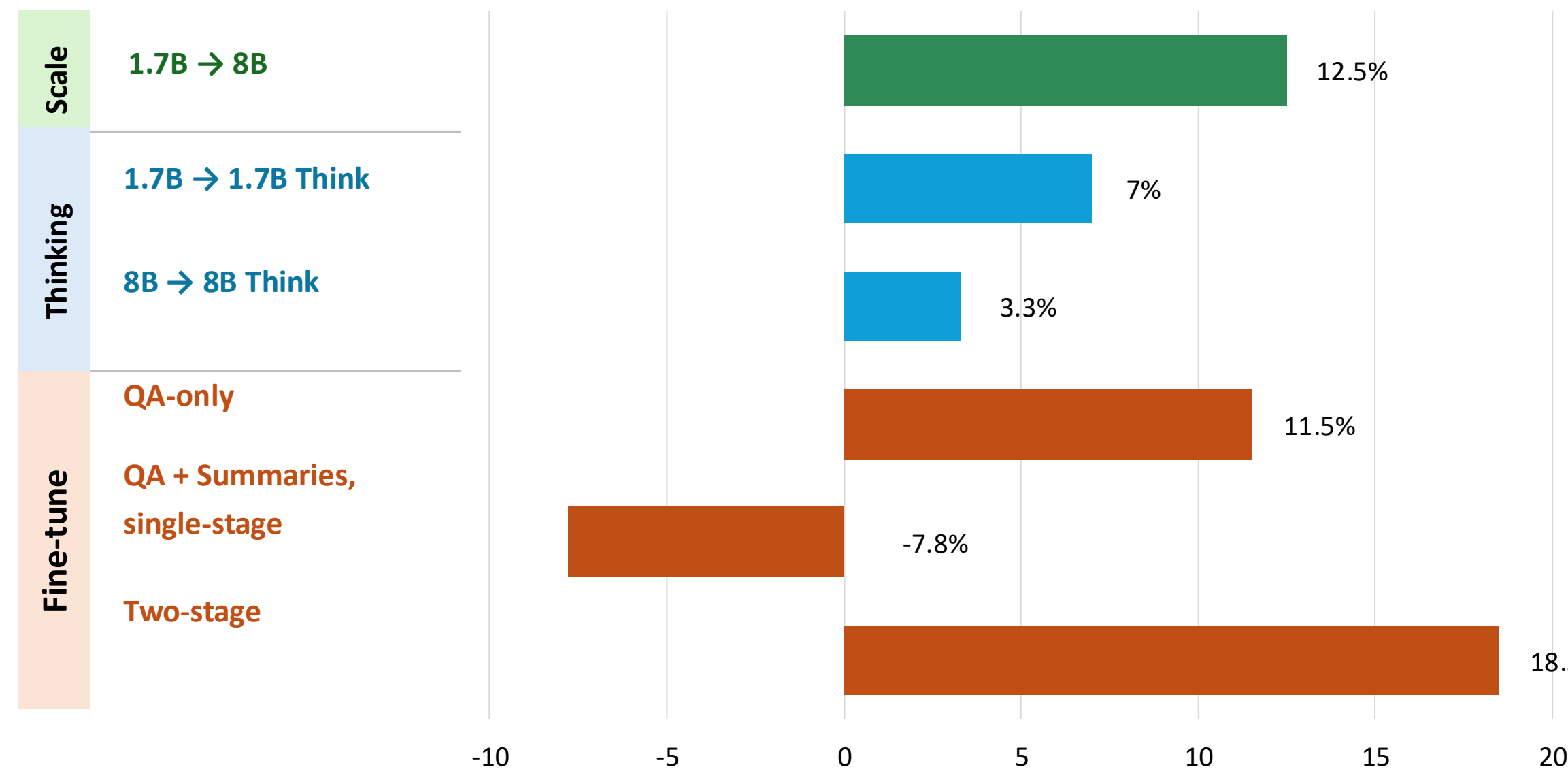


Figure 6. Two-stage fine-tuning gives the largest gain over Qwen3-1.7B Base (+18.5 points), exceeding the gains from scaling to 8B (+12.5 points) or enabling thinking mode.

Limitations & Future Work

- Gemini both generated the benchmark and was evaluated on it; its 78.2% may partly reflect stylistic familiarity with its own question phrasing rather than domain knowledge alone.
- Multiple-choice accuracy is objective and reproducible, but it does not measure explanation quality, citation use, uncertainty, or robustness to paraphrase.
- Some factual and quantitative questions are paper-specific, but all models are evaluated closed-book.
- Only Qwen3-1.7B was fine-tuned, so it remains unclear whether the same recipe generalizes to larger open models or other model families.

Next step: Build a retrieval-augmented methylation QA assistant that combines fine-tuning for domain fluency and answer-format alignment with retrieval for exact paper-level grounding.

Conclusion

Small open models can be meaningfully adapted for specialized biomedical QA. On this 400-question DNA methylation benchmark, Gemini 2.5 Flash performs best at 78.2%, but a two-stage fine-tuned Qwen3-1.7B model reaches 74.3% and outperforms Qwen3-8B Thinking.

Final takeaway: For narrow scientific domains, the structure of fine-tuning can be as important as model scale. A promising methylation QA assistant should combine fine-tuning for domain fluency with retrieval for factual grounding in the source literature.

